CLASS 10 SCIENCE: CHEMICAL REACTIONS & EQUATIONS

Practice Worksheet (Unsolved)

SECTION A: MULTIPLE CHOICE QUESTIONS

Question 1: $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$
Identify the substance oxidized in the above equation.

(a) $MnCl_2$ (b) HCl (c) H_2O (d) MnO_2

Question 2: Electrolysis of water is a decomposition reaction. The mole ratio of hydrogen and oxygen gases liberated during electrolysis of water is:

(a) 1 : 1 (b) 2 : 1 (c) 4 : 1 (d) 1 : 2

Question 3: $Pb + CuCl_2 \rightarrow PbCl_2 + Cu$
The above reaction is an example of:

(a) combination (b) double displacement (c) decomposition (d) displacement

Question 4: When green coloured ferrous sulphate crystals are heated, the colour of the crystal changes because:

(a) it is decomposed to ferric oxide (b) it loses water of crystallisation (c) it forms SO_2 (d) it forms SO_3

Question 5: $2HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + 2H_2O$ is an example of:
(i) displacement reaction (ii) double displacement reaction
(iii) neutralisation reaction (iv) combination reaction.

(a) (i) and (ii) (b) (ii) and (iii) (c) (iii) and (iv) (d) (i) and (iv)

Question 6: What is observed when a solution of potassium iodide is added to silver nitrate solution?

(a) No reaction takes place (b) White precipitate of silver iodide is formed (c) yellow precipitate of AgI is formed (d) AgI is soluble in water

Question 7: Identify 'x', 'y' and 'z' in the following balanced reaction:
$x\ Pb(NO_3)_2(s) \rightarrow y\ PbO(s) + z\ NO_2(g) + O_2(g)$

(a) 2, 4, 2 (b) 2, 2, 4 (c) 2, 4, 4 (d) 4, 2, 2

Question 8: Oxidation involves:
(i) gain of electron (ii) loss of electron
(iii) addition of oxygen or electronegative element (iv) removal of hydrogen or electropositive element

(a) (i), (ii), (iii) (b) (ii), (iii), (iv) (c) (i), (iii), (iv) (d) (i), (ii), (iv)

Question 9: Which of the following statements about the given reaction are correct?
$3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$

(i) Iron metal is getting oxidised (ii) Water is getting reduced
(iii) Water is acting as reducing agent (iv) Water is acting as an oxidising agent

(a) (i), (ii) and (iii) (b) (iii) and (iv) (c) (i), (ii) and (iv) (d) (ii) and (iv)

Question 10: Barium chloride on reacting with ammonium sulphate forms barium sulphate and ammonium chloride. Which of the following correctly represents the type of the reaction involved?
(i) Displacement reaction (ii) Precipitation reaction
(iii) Combination reaction (iv) Double displacement reaction

(a) (i) only (b) (ii) only (c) (iv) only (d) (ii) and (iv)

SECTION B: ASSERTION-REASON TYPE QUESTIONS

DIRECTION: Each of these questions contains an Assertion (A) followed by Reason (R). Read them carefully and select the correct option from the choices below:
(a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are correct, but Reason is NOT the correct explanation of Assertion.
(c) Assertion is correct but Reason is incorrect.
(d) Assertion is incorrect but Reason is correct.

Question 1: Assertion (A): Photosynthesis is considered as an endothermic reaction.
Reason (R): Energy gets released in the process of photosynthesis.

Question 2: Assertion (A): Decomposition of vegetable matter into compost is an example of exothermic reactions.
Reason (R): Exothermic reactions are those reactions in which heat is evolved.

Question 3: Assertion (A): When HCl is added to zinc granules, a chemical reaction occurs.
Reason (R): Evolution of a gas indicates that the chemical reaction is taking place.

Question 4: Assertion (A): Calcium carbonate when heated gives calcium oxide and carbon dioxide.
Reason (R): On heating calcium carbonate, decomposition reaction takes place.

Question 5: Assertion (A): Chemical reaction changes the physical and chemical properties of a substance.
Reason (R): Chemical change involves a change in the chemical composition of matter, and a new substance is formed.

Question 6: Assertion (A): AgBr is used in photographic and X-ray film.
Reason (R): AgBr is photosensitive and changes to Ag and bromine in presence of sunlight and undergoes decomposition reaction.

Question 7: Assertion (A): Magnesium ribbon keeps on burning in atmosphere of nitrogen.
Reason (R): Magnesium reacts with nitrogen to form magnesium nitrides and this reaction is combination reaction.

Question 8: Assertion (A): Zinc reacts with sulphuric acid to form zinc sulphate and hydrogen gas and it is displacement reaction.
Reason (R): Zinc reacts with oxygen to form zinc oxide.

Question 9: Assertion (A): $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$ is redox reaction.
Reason (R): MnO_2 oxidises HCl to Cl_2 and gets reduced to $MnCl_2$.

Question 10: Assertion (A): White silver chloride turns grey in sunlight.
Reason (R): Decomposition of silver chloride in presence of sunlight takes place to form silver metal and chlorine gas.

SECTION C: CASE STUDY QUESTIONS

Read the following passage and answer the questions that follow:

Reactions in which one element takes place of another element in a compound, are known as displacement reactions. In general, a more reactive element displaces a less reactive element from its compound. In all single displacement reactions, only one element displaces another element from its compound. The displacement reaction between iron (III) oxide and powdered aluminium produces so much heat that iron metal obtained is in molten form.

Question (i): Copper displaces which of the following metals from its salt solution?

(a) $ZnSO_4$ (b) $FeSO_4$ (c) $AgNO_3$ (d) $NiSO_4$

Question (ii): When zinc reacts with dilute sulphuric acid, the gas evolved is:

(a) red in colour and have a sweet smelling
(b) green in colour and have a foul smell
(c) colourless, odourless and burns with a pop sound
(d) colourless, pungent smelling and burns with a pop sound

Question (iii): When dry hydrogen is passed over a heated oxide of metal X using standard apparatus, a reddish-brown residue is obtained. The reddish-brown residue could be:

(a) copper (b) lead (c) silver (d) zinc

Question (iv): Which of the following reactions is a displacement reaction?

(a) $CaO + H_2O \rightarrow Ca(OH)_2$ (b) $MgCO_3 \rightarrow MgO + CO_2$
(c) $Mg + CuSO_4 \rightarrow MgSO_4 + Cu$ (d) $H_2 + Cl_2 \rightarrow 2HCl$

Question (v): When dilute hydrochloric acid is added to granulated zinc placed in a test tube, the observation made is:

(a) the surface of the metal turns shining (b) the reaction mixture turns milky
(c) greenish yellow gas is evolved (d) the colourless and odourless gas evolves with a pop sound.

SECTION D: SHORT ANSWER TYPE QUESTIONS

Question 1: List four observations that help us to determine whether a chemical reaction has taken place. [Board Term-I, 2012]

Question 2: What is observed when carbon dioxide gas is passed through lime water:
(i) For a short duration
(ii) For long duration?
Also, write the chemical equations for the reactions involved. [Board Term I, 2016]

Question 3: Write the balanced chemical equation for the following reaction and write the name of the reaction:
Barium Chloride + Aluminium Sulphate → Barium Sulphate + Aluminium Chloride

Question 4: Why do we store silver chloride in dark coloured bottles? Explain in brief. [CBSE 2010]

Question 5: Define a combination reaction. Give one example of a combination reaction which is also exothermic.

Question 6: What happens when an iron nail is put inside a copper sulphate solution? Write a reaction with observation.

Question 7: Name the raw materials that are required for the manufacture of washing soda by Solvay process. Describe the chemical reactions involved in the process. [2007]

Question 8: (i) State the law which is followed in balancing a chemical equation.
(ii) Balance the following chemical equation: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$ [Board Term-I, 2012]

Question 9: A white solid when dropped in water produces a hissing sound. What may the solid be? Give the chemical reaction for the above. Name the product formed. [2010]

Question 10: When hydrogen gas is passed over heated copper (II) oxide, copper and steam are formed. Write the balanced chemical equation with physical states for this reaction. State what kind of chemical reaction this is. [Board Term-I, 2015]

SECTION E: LONG ANSWER QUESTIONS

Question 1: (a) 2g of ferrous sulphate crystals are heated in a dry boiling tube.
(i) List any two observations.
(ii) Name the type of chemical reaction taking place.
(iii) Write the chemical equation for the reaction.
(b) Write the chemical equation of the reaction in which the following changes have taken place with an example of each:
(i) Change in colour (ii) Change in temperature (iii) Formation of precipitate

Question 2: Raju made the following observations during the reaction of some metals with dilute hydrochloric acid.
(a) Silver metal does not show any change.
(b) The temperature of the reaction mixture rises when aluminium (Al) is added.

(c) The sodium metal reaction is highly explosive.
(d) Some gas bubbles are seen when lead (Pb) is reacted with the acid.
Explain these observations giving suitable reasons.